Automation Engine Troubleshooting

Guide	Run Book Automation Engine
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This is a troubleshooting guide that's meant to help in determining problems or issues related to Automation Engine performance.

- Other Articles
- How does the Automation Engine work?
- Why isn't my Action firing?
- How can I determine what the Automation Engine is doing?
- The Automation Engine is backed up. How do I clear it out?
- What is causing the Automation Engine to be slow?
- How many Automation Engine processes should I expect to be running on the CDB?
- When are variable values populated for a triggered RBA?
- Can a cleared RBA execute for a clearing event record before the respective "active" RBA executes for that event record?
- Why isn't the device reported the one I expect from the RBA variable (%_parent_id)?
- Why do RBA Actions fail to execute from System events in 11.1.x?
- Why do all RBA snippet actions fail for certain devices while not for other devices?
- How does the Automation Engine know if an external ticket has already been created for an event in SL1?
- Notification log shows "Message:Action ## result exceeded the maximum size limit. Result: None". What does this mean?
- Notification log shows "ERROR rba_executor.queue_on_collector ## Unable to send task to Collector. Action object exceeds the max limit while running Automation Policy ## Action ##". What does this mean?
- Queries for Automation Engine Metrics
  - Current Automation Engine Cycle
  - Current Load vs Capacity on Automation Engine
  - Popular Automation Policies (e.g. defined sequences of actions) (and how long they take)
  - Current Automation Engine Policies Triggered per Hour
  - Current Automation Engine Actions per Hour
  - Popular Automation Actions (and how long they take)
  - How many tasks are queued up for each RBA queue? (11.2.0+)
- What about the Debug Logs?
- DEBUG message meanings:

Other Articles

This is another guide with specific diagnosis and mitigation procedures: Troubleshooting the Automation Engine: main article. It walks you through how to analyze the RBA action queuing and execution timing, which can identify which RBAs are slowing the system down.

How does the Automation Engine work?

Recommend that you read the overview on the Automation Engine.

Why isn't my Action firing?

- Identify whether this is a scenario where the anticipated Automation Policy didn't run or whether a given action within that policy execution did not execute.
- If the event is still active on the Event Console the Notifier Log can be viewed to show Action execution **against the event** otherwise these may be found in the CDB via

```
SELECT * FROM `master_logs`.`notifier_log`;
```

> ⚠️ Time point schedule-based Run Book Automation does not have the context of an event record and therefore does **not** get recorded to `master_logs`.`notifier_log`; the execution are recorded, along with all other RBA executions, to the respective automation metrics tables

- IF the Automation Policy didn't execute:
  - Verify the Automation Policy is enabled
  - Make sure the entity against which the event is aligned is within the Automation Policy criteria.
  - Make sure the Event Policy for the event is aligned within the Automation Policy criteria

    - Take note of the Organization alignment. (If the RBA policy is aligned to OrganizationA, but the entity where the event fired is not within that Org, the Automation Policy will not evaluate to execute.)
- IF a given action within the Automation Policy did not execute:
    - Make sure the action is aligned to the Automation Policy
    - Check the action context - is this set to run on the CDB or CU?

How can I determine what the Automation Engine is doing?

1. Version >= 8.2.0, the Automation Engine will send a warning when it has been unable to load tasks for a given period of time.
    a. By default, this is 180 seconds (This is configurable in `master.system_settings_automation_engine`).
    b. Version >= 8.4.0, this message will also include the priority of the tasks that it has been unable to load.
    c. Search for the word *automation* in either the Event Console or System Logs.
2. The `master_logs.notifier_log` will give an human readable summary of the completed actions, whether they were successful and what the results were (if any).
    - **Note:** If a Snippet action was unsuccessful, this may have been because of an error, or simply because the Snippet took too long.
        - *Version < 8.1.5:* The Snippet Timeout is 60 seconds.
        - *Version >= 8.1.5:* The Snippet Timeout is 300 seconds, by default. It is configurable in `master.system_settings_automation_engine`.
3. You can find out how an individual action is performing by looking at the appropriate metrics table.
    - *Version < 8.4.0:* The `master_logs.automation_action_runs` table will give timing, CPU and memory statistics, *but only for Snippet actions*. Unfortunately, there is no way to determine which Snippet applies to a given metric and there are no other metrics available without turning on the debug logs for the Automation Engine.
    - *Version >= 8.4.0:* The `master_logs.automation_action_metrics` will give timing information for all actions and ~~CPU and memory for Snippet actions~~.
4. Version >= 8.4.0, you can find out how various event/policy combinations are doing by investigating the `master_logs.automation_policy_metrics` table.
    - The **Event Trigger Time** is the theoretical time when an event triggered (or re-triggered) the Automation Engine to look for associated Automation Policies.
    - The **Task Loaded Time** is when the Job Feeder started to load tasks.
        - The Job Feeder will called every 10 seconds during lighter loads and less frequently when the system is more heavily loaded.
    - The **Task Queued Time** is when the automation task was placed on the Task Queue.
    - The **Task Started Time** is when an RBA Executor picked up the Task off of the Task Queue.
        - Note: For Tasks that run on a Collector, this corresponds to the time when the Task was shipped off to a Collector to Run, not when the Collector started to run it.
    - The **Task Completed Time** is when the RBA Executor finished running the Task.

The Automation Engine is backed up. How do I clear it out?

For example, receiving System events stating "The automation engine is still processing ##### tasks and has been unable to retrieve new runbook actions for HH:mm:ss"

There is no simple way to "clear" the Automation Engine. If the Automation Engine is simply restarted (without modifying anything), it will immediately load itself back into a similar state to when it was shut down. The following options are available.

- Disable some Action Policies or Automation Policies temporarily that are causing the slowdown.
    - **Note:** Version >= 8.4.0, you can change the priority of important Automation Policies so that the bottleneck is not affecting all tasks. Or you can lower the priority of the bottleneck causing automation policies, so other Automation Tasks will be executed sooner.
- Remove some of the Active or Cleared Events that are causing so many tasks to be executed.
- (Version >= 8.8.0) Remove some of the scheduled RBA tasks in `master_automation`.`services`
    - type "1" = scheduled
    - type "2" = ad hoc
- If it is truly a matter of volume, in version >= 8.1.5, you can add more executors to process the backlog (by configuring it in `master.system_settings_automation_engine`).
    - **Note:** If a downstream system (or the EM7 Database) is being slow to respond to a Snippet or Action, increasing the number of executors may make the problem *worse,* not better.

After you have done one or more of the above, restarting the Automation Engine should allow it to start processing through the backlog faster.

What is causing the Automation Engine to be slow?

We have seen four issues that will cause a delay from when the message is received by EM7 to when the Automation Engine is able to complete an Action based on that message/event.

1. There are a large number of tasks that the Automation Engine needs to process. This is caused by one of two things:
    a. There was a surge in the amount of active and/or cleared events that EM7 is receiving. Check the `master_events.events_active` or `master_events.events_cleared` for an increase in the amount of events over the given few hours.
    b. There is an Automation Policy that applies to too many events (or is repeating too often). In 8.4.0, check the `master_logs.automation_policy_metrics` table if a policy is appearing too often in the previous few hours.
2. There is an action (or a set of actions) that are running slowly.
    a. Typically, this is a Snippet. The suggestion is to look at the metrics tables (`master_logs.automation_action_runs` or `master_logs.automation_action_metrics`) to determine if it is a Snippet and how long that Snippet is taking. If that snippet has a log, look for errors or timing issues in it.
    b. Often, a downstream system (either a Database or Remote Server) or the EM7 Database itself is slow to respond to requests and that is "slowing" the Automation Engine down. Check if that Snippet / DB Query has some logs to see how long that request is taking. If verified, the recommendation is to disable that Action Policy that is connecting to that system/service, until it recovers.
3. There are a significant amount of Automation Tasks that need to be run on a Collector.
    - Needing to transfer the Automation Task back and forth from a Collector can significantly increase the time that it requires to run. Determine if the particular action(s) need to be executed on a Collector. If they do, determine if you can restrict the number of Events that the Automation Policy applies to (or in version 8.4.0, change the priority of those policies).
4. Events aren't getting to the Automation Engine in a reasonable amount of time, which is actually an Event Engine problem.

- If there are no signs of a performance problem in the Automation Engine, then this is the likely culprit. Verify that the Event Engine is performing okay and that the incoming queues aren't backed up. You will need to check the message tables on the appropriate collector (`net_snmp.notifications`, `in_syslog.messages`, `in_dynamic.messages`, etc).

How many Automation Engine processes should I expect to be running on the CDB?

On a CDB one can expect the following count of *`usr/bin/python2.7 /opt/em7/backend/automation_eng.py`* process running:

> 1 (Automation Engine main process) + 1 (process for loading/adding Schedule tasks) + 1 (process for loading/adding User Initiated tasks) + X (# Executors) + Y (# of RBA snippets executing where 0 <= Y <= X)

When are variable values populated for a triggered RBA?

Event state data that populates these variables is captured at the time the RBA task is triggered and used later when the RBA is executed. The only thing we re-check before executing the RBA action is check whether action is still enabled.

Can a cleared RBA execute for a clearing event record before the respective "active" RBA executes for that event record?

Yes. If the event is generated and then cleared immediately, and the "cleared event" RBA utilizes "and no time elapsed" criteria, as the Automation Engine handles "cleared" event execution first in each cycle then the clearing RBA may execute before the respective "active" execution was able to trigger.

Why isn't the device reported the one I expect from the RBA variable (%_parent_id)?

This variable populated via the following query behind-the-scenes:

```
SELECT p_did FROM aster_dev.V_entity_relationships WHERE c_did=<your_device_id>
```

Accurate as of 11.2.0, the `V_entity_relationships` table does not have an imposed ordering and if there are multiple parent-child relationships defined for the device the one returned may conflict with expectations.

[UPDATE] This was changed

Why do RBA Actions fail to execute from System events in 11.1.x?

Please refer to EM-49372 - RBA actions fail to execute from System Events in 11.1.x CLOSED for more information on this issue. There is a workaround available, as well as a more permanent solution offered in more recent releases.

Why do all RBA snippet actions fail for certain devices while not for other devices?

This *could* be an instance of a corrupt Dynamic App credential that's aligned to the device. Still valid as of 11.3.0, during execution of an RBA snippet triggered from a device-based event, the Automation Engine loads all Dynamic App aligned credentials from that device as parameters. A corrupt credential (whether entirely broken or containing a huge amount of data) can cause such impact. (See related cases
EM-15765 - DE81502 [Defect] Credentials created via the API are "corrupt" CLOSED &
EM-53275 - Service Now Runbook Automation failing on Generic Linux Devices CLOSED)

How does the Automation Engine know if an external ticket has already been created for an event in SL1?

The Automation Engine evaluates whether the event record has an entry for `ext_ticket_ref` and whether `master_biz.ticket_external_requests` has an entry for the event.

Notification log shows "Message:Action ## result exceeded the maximum size limit. Result: None". What does this mean?

This error means that the action result exceeds 65354 bytes; the maximum size an actual result can handle. Subsequent actions within the RBA sequence will execute, however if they depend on the content of the large result they may not operate correctly. The ID of the Run Book Action responsible is contained within the message and should be reviewed and adjusted to reduce the result returned.

Technical explanation: *Since Linux 2.6.11, the pipe capacity is 16 pages (i.e., 65,536 bytes in a system with a page size of 4096 bytes).*

Notification log shows "ERROR rba_executor.queue_on_collector ## Unable to send task to Collector. Action object exceeds the max limit while running Automation Policy ## Action ##". What does this mean?

This error means that the action snippet + previous stored results from prior actions within this Run Book Automation policy exceeds 65354 bytes and cannot be sent to the Collector for execution. Subsequent actions within the RBA sequence

will execute, however if they depend on the content of the large result they may not operate correctly or if they too must be sent to a Collector for execution.

Technical explanation: *Since Linux 2.6.11, the pipe capacity is 16 pages (i.e., 65,536 bytes in a system with a page size of 4096 bytes).*

Queries for Automation Engine Metrics

Current Automation Engine Cycle

How often the job feeder is running. If the automation engine is "slow", often times it's because these cycles take a long time.

```
SELECT task_loaded_time, count(*) as tasks_loaded
  FROM master_logs.automation_policy_metrics
 WHERE task_loaded_time > (NOW() - interval 10 hour)
 GROUP BY task_loaded_time
 ORDER BY task_loaded_time desc
```

Current Load vs Capacity on Automation Engine

What has been the load on the RBA Engine over the last hour? These results can be compared against estimated max capacity for that hour calculated by (1 hour * number of executors * 60 to convert to minutes * 60 to convert to seconds [ex. with 5 executors you have an estimated max capacity of 18000 seconds of execution time])

```
SELECT count(*) as automations_ran, avg(task_duration) as avg_time, max(task_duration) as max_ti
   FROM master_logs.automation_policy_metrics
  WHERE task_started_time>date_sub(now(), interval 1 hour)
```

Popular Automation Policies (e.g. defined sequences of actions) (and how long they take)

Which Automation policies are running the most often and how long they are taking. These are the ones you'll probably want to investigate further as to what they are doing.
Note: You'll probably want to limit this query by some number of rows.

```
SELECT ap_id, count(*) as automations_ran, avg(task_duration) as avg_time, max(task_duration) as
  FROM master_logs.automation_policy_metrics
 WHERE task_completed_time > (NOW() - interval 1 day)
 GROUP BY ap_id
 ORDER BY automations_ran desc, avg_time desc
```

Current Automation Engine Policies Triggered per Hour

Throughput of automation policies for the automation engine. Useful for seeing if there are spikes in automations triggered or long running policies (e.g. sequence of actions). This could be misleading if the automation policy had a long execution and crossed and hour boundary.

```
SELECT dayofmonth(task_started_time) as day, hour(task_started_time) as hr, count(*) as automati
  FROM master_logs.automation_policy_metrics
 WHERE task_started_time > (NOW() - interval 24 hour)
 GROUP BY day, hr
 ORDER BY day desc, hr desc
```

Current Automation Engine Actions per Hour

Throughput of actions for the automation engine. Useful for seeing if there's spikes of actions. Note that this can be misleading, a bunch of shorter actions may be the same as a lower number of longer ones).

```
SELECT dayofmonth(end_time) as day, hour(end_time) as hr, count(*) as actions_ran, avg(duration)
  FROM master_logs.automation_action_metrics
 WHERE end_time > (NOW() - interval 1 day)
 GROUP BY day, hr
 ORDER BY day desc, hr desc
```

Popular Automation Actions (and how long they take)

Which actions are running the most often and how long they are taking. These are the ones you'll probably want to investigate further as to what they are doing.
Note: You'll probably want to limit this query by some number of rows.

```
SELECT action_id, count(*) as actions_ran, avg(duration) as avg_time, max(duration) as max_time,
  FROM master_logs.automation_action_metrics
 WHERE end_time > (NOW() - interval 1 day)
 GROUP BY action_id
 ORDER BY actions_ran desc, avg_time desc
```

How many tasks are queued up for each RBA queue? (11.2.0+)

With multiple queues with RBA Automation, how do I check the latest queue sizes (with the current queue size only being in memory?) The following query will provide the latest recorded queue sizes.
Note: Active queue sizes are captured every 1 minute

```
select * from `master_logs`.`automation_queue_sizes` ORDER BY id desc LIMIT 1;
```

What about the Debug Logs?

EM7 Core: Automation Engine for CDB processing

EM7 Core: Automation Agent for Collector execution

Automation Engine **must** be restarted for debug to take effect

DEBUG message meanings:

```
"2019-08-11 13:04:17,116","DEBUG","20190","67","policy_checks","150","This event (27887) is not
```

Meaning: Event ID 27887 was evaluated for the RBA Policy (which would be mentioned in the proceeding log message) is not the parent event (event ID 27885) in the event group, and the RBA Policy is **not** configured for "Trigger on child rollup"